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Turbocharger Training for Fixed Geometry Products

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Introduction

The turbocharger is a critical part of the air handling system and engine performance.

The components within the turbocharger are often more sensitive than other engine components. This can cause the turbocharger to show early signs of an engine and/or system issue.

The intent of this training is to help explain the function of the turbocharger, the turbocharger's system interaction, what can cause failures, and other aspects of the turbocharger.

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- Turbocharger Familiarization
- Bearing System
- How a Turbocharger Seals Oil
- System Interaction
- Wheel Fatigue
- Summary

Turbocharger Familiarization

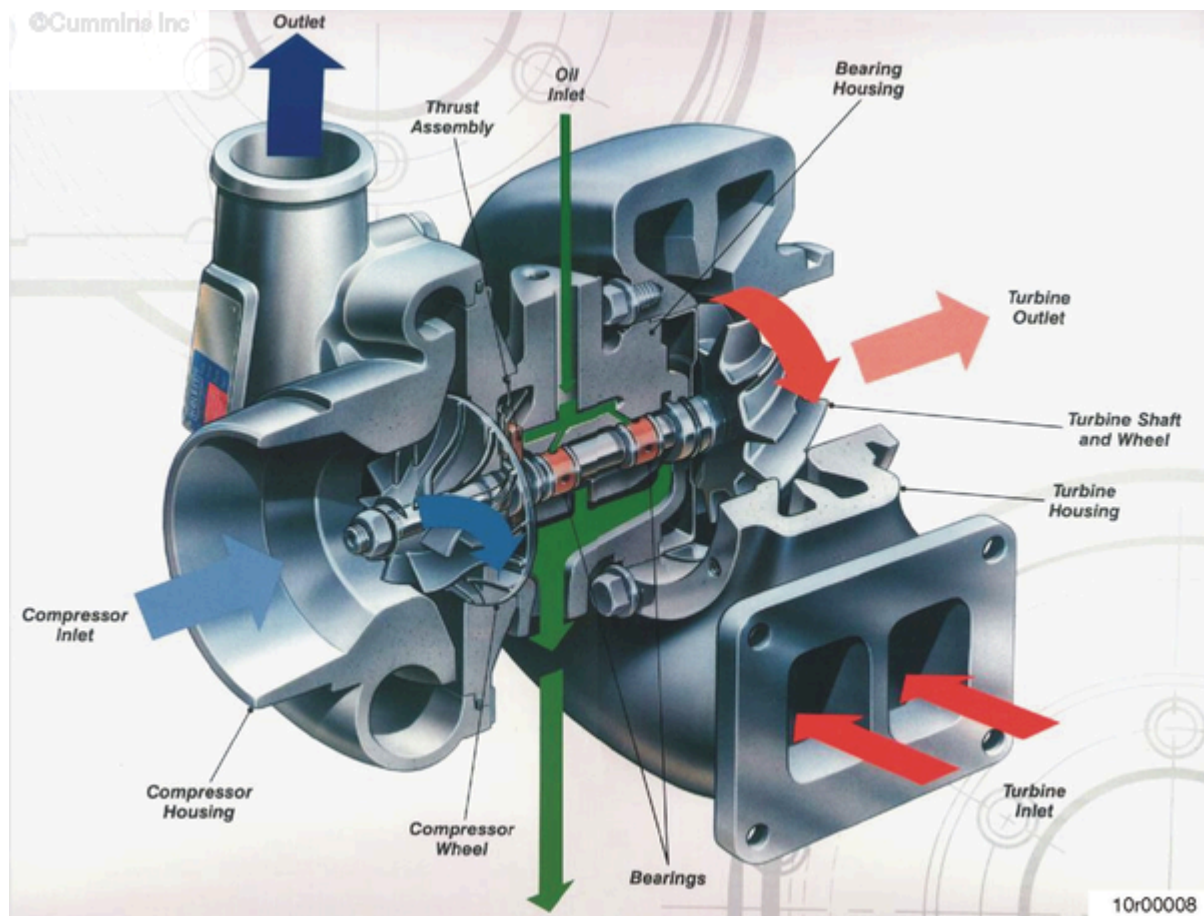


Figure 1: Fixed Geometry Turbocharger

Figure 1 shows typical turbocharger components, intake air flow, exhaust gas flow, and lubricating oil flow.

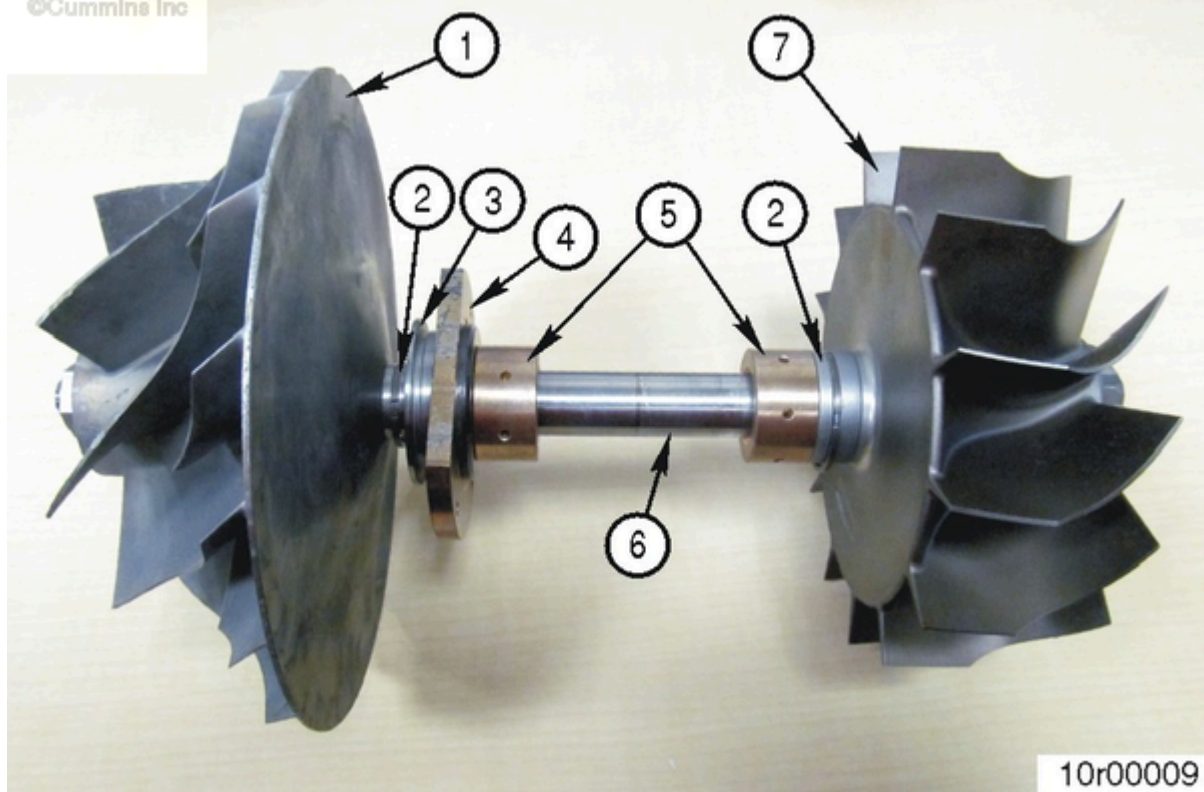


Figure 2: Rotor assembly.

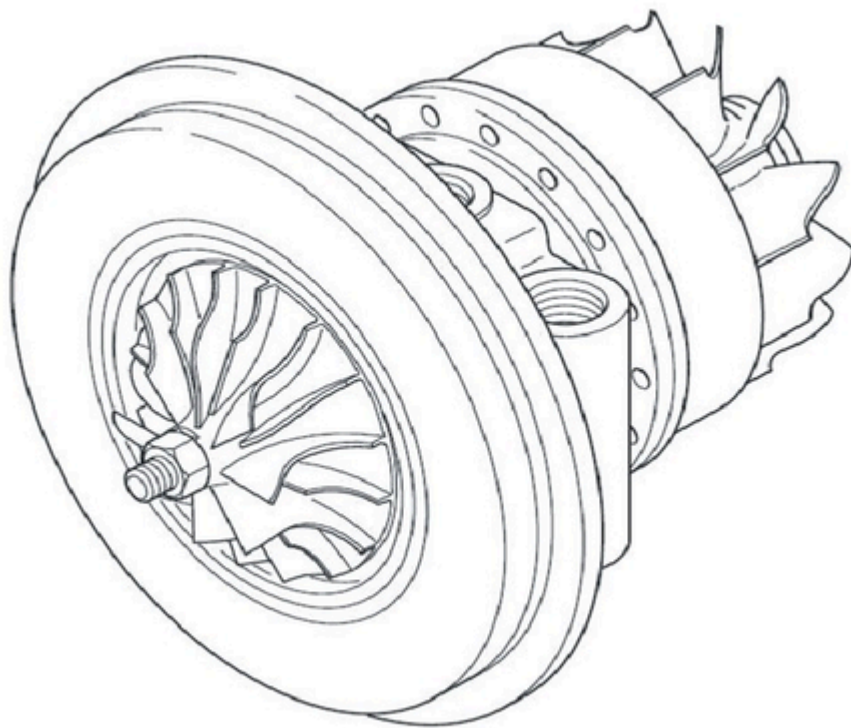


Figure 3: Bearing housing assembly.

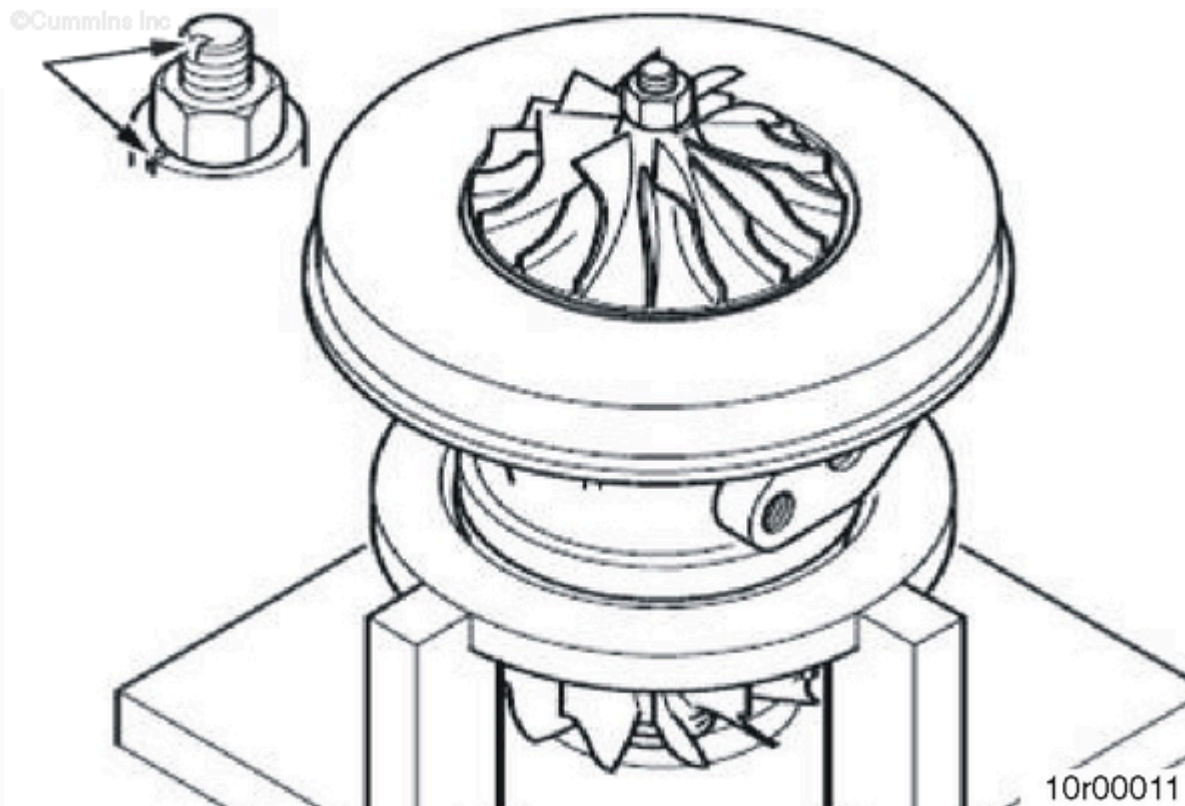


Figure 4: Alignment mark indicates where turbine and impeller wheel are balanced.

Figure 2 shows the following turbocharger components:

1. Compressor wheel
2. Split ring seals
3. Thrust collar
4. Thrust bearing
5. Journal bearings
6. Shaft
7. Turbine wheel.

Figure 2 and Figure 3 show the bearing housing assembly. The assembly contains the following:

- Journal Bearings - These are free floating bearings that rotate on a film of oil. They help to control radial motion.
- Thrust Bearing - This is a lubricated component to lubricate the oil slinger and thrust collar, as well as help with thrust loading.
- Thrust Collar - Used to help control thrust loading (axial motion).
- Oil Slinger - A rotating component used to help keep oil within the bearing housing.
- Heat Shield - Used to help protect the bearing system from the high heat within the turbine housing.

Rotor assemblies are a balanced assembly. The turbine shaft and wheel **must** be aligned properly to the impeller wheel to have the correct balance. This alignment is marked for assembly and disassembly, as shown in Figure 4.

Bearing System

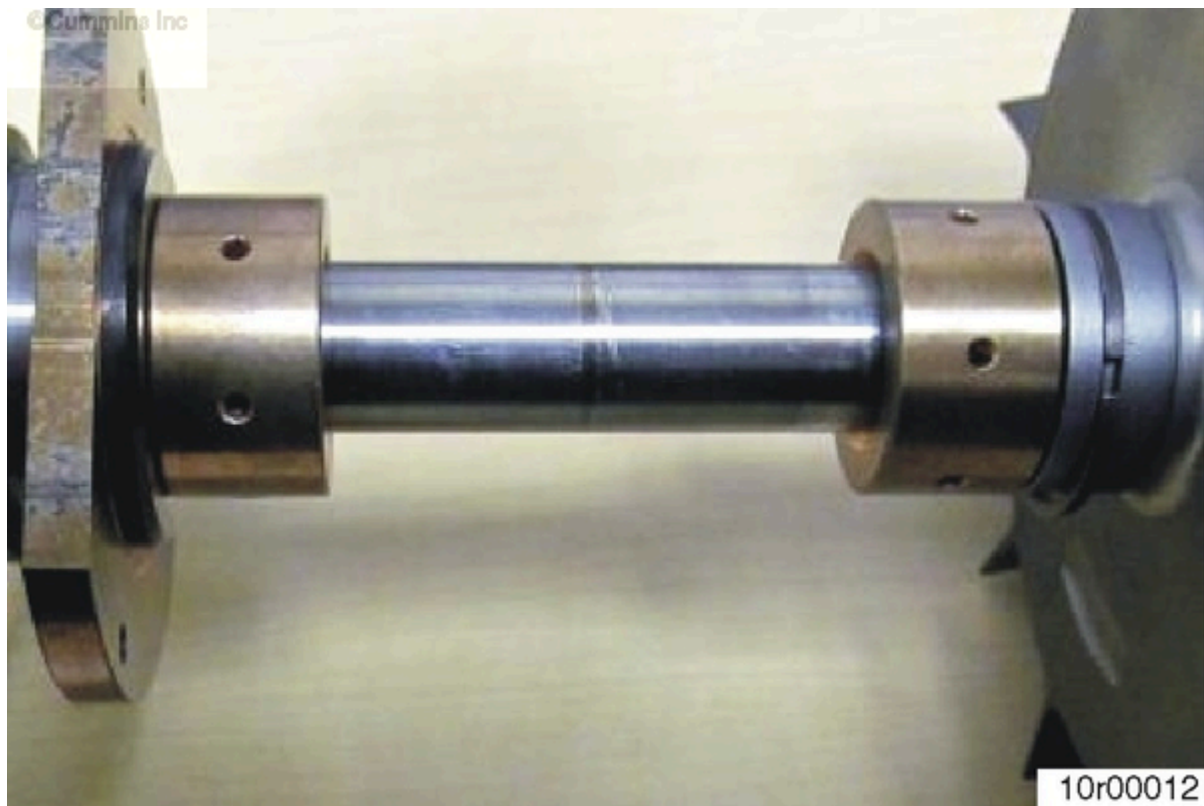


Figure 5: Journal bearings shown on shaft.

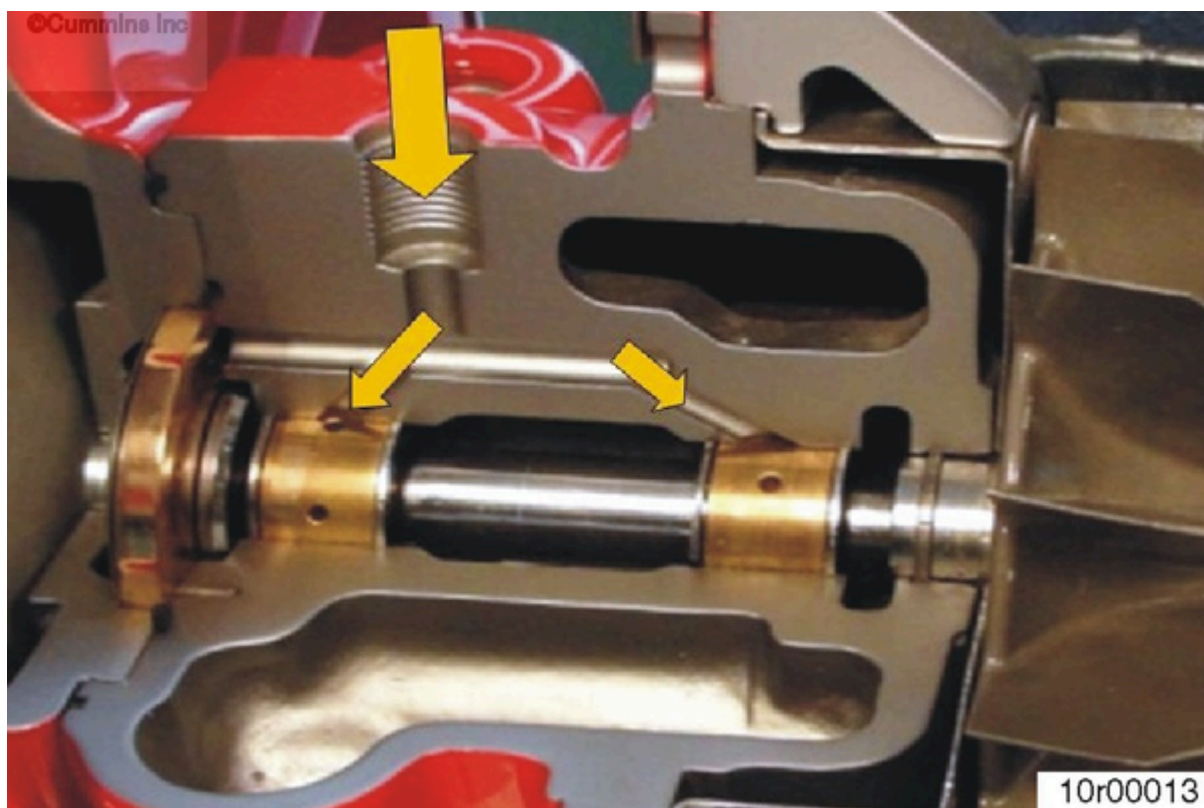


Figure 6: Oil supply to bearings.

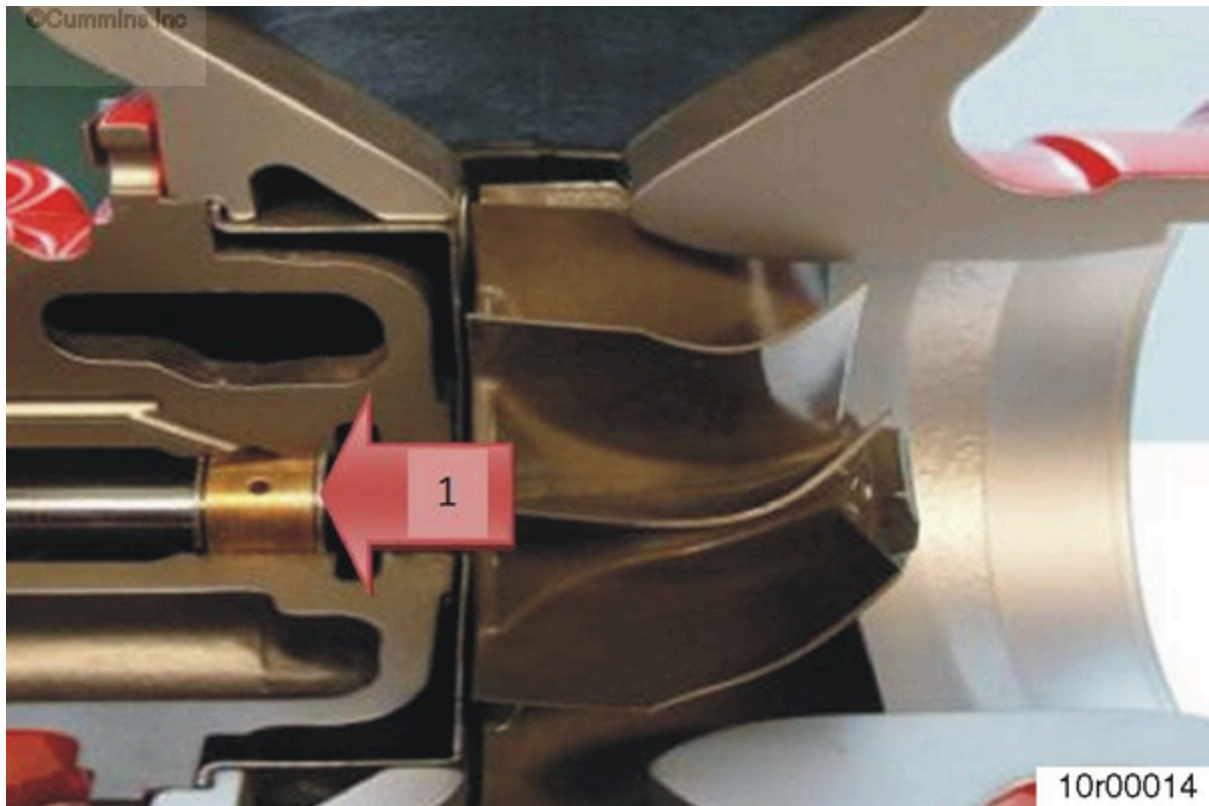


Figure 7: Coked oil at the turbocharger bearing housing.

There are three main causes of bearing failures:

- Overheated oil
- Oil contamination
- Insufficient lubrication

A bearing failure can cause the impeller or turbine wheel to contact its housing and result in a turbocharger failure.



Figure 8: Turbocharger bearings with coked oil burnt onto them.



Figure 9: Impeller/rotor shaft with coked oil burnt onto it.

Overheated Oil

Overheated, or coked, oil is generally caused by hot shutdowns*, excessive high exhaust temperatures, or high engine oil temperatures.

* A hot shutdown is when the engine is shut off after being run hard. When the engine is **not** allowed a period of idling after being run hard, the turbine housing is still very hot. With the coolant and oil systems no longer operating, heat transfer can occur from the turbine housing. The oil sitting at the journal bearings can increase in temperature and turn into a hard, coked, oil, as shown in Figure 8 and Figure 9.

Coked oil deposits are abrasive and can wear away the bearing material.

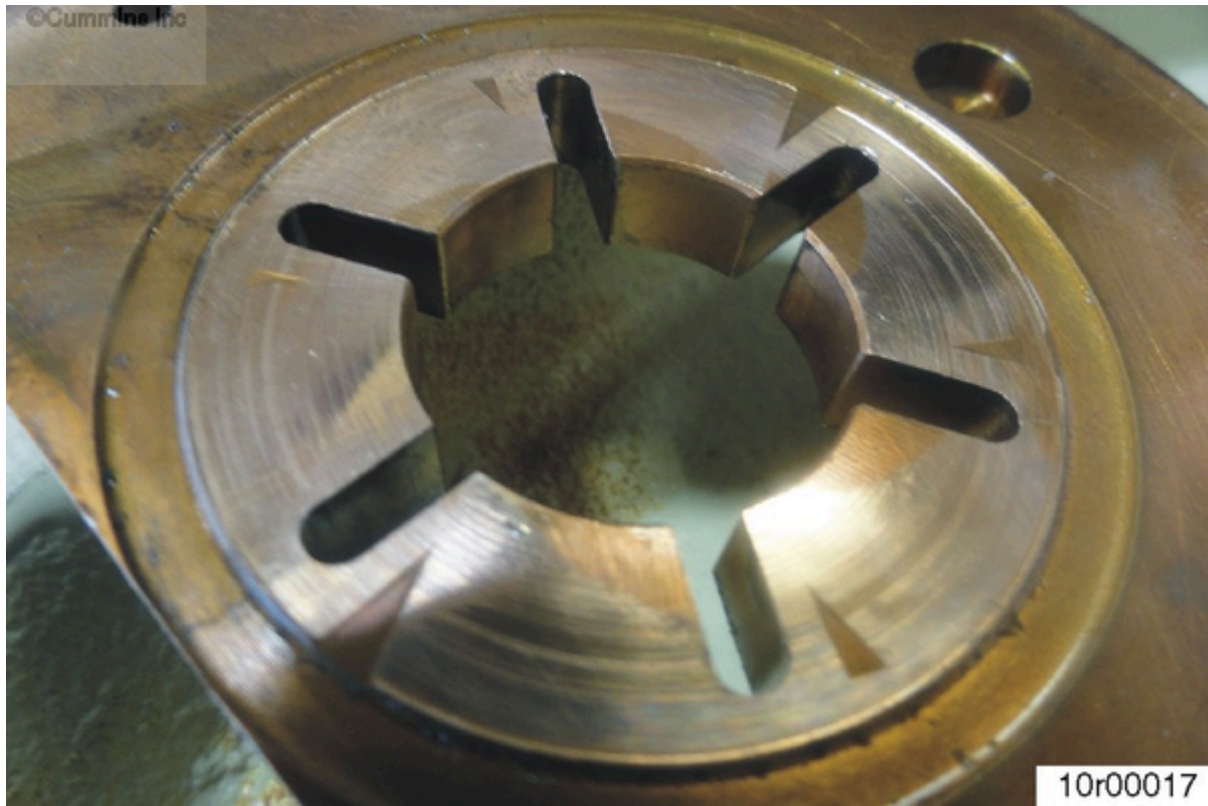


Figure 10: Turbocharger thrust bearing wear.



Figure 11: Turbocharger bearing damage. Debris in the oil can score and wear the journal and thrust bearings.

Oil Contamination:

Figure 10 and Figure 11 show damage from debris in the lubricating oil.

Debris in the oil can wear the bearing material and eventually lead to a bearing failure. If a bearing failure is suspected, the oil should be checked for debris.



Figure 12: Damaged thrust bearing.

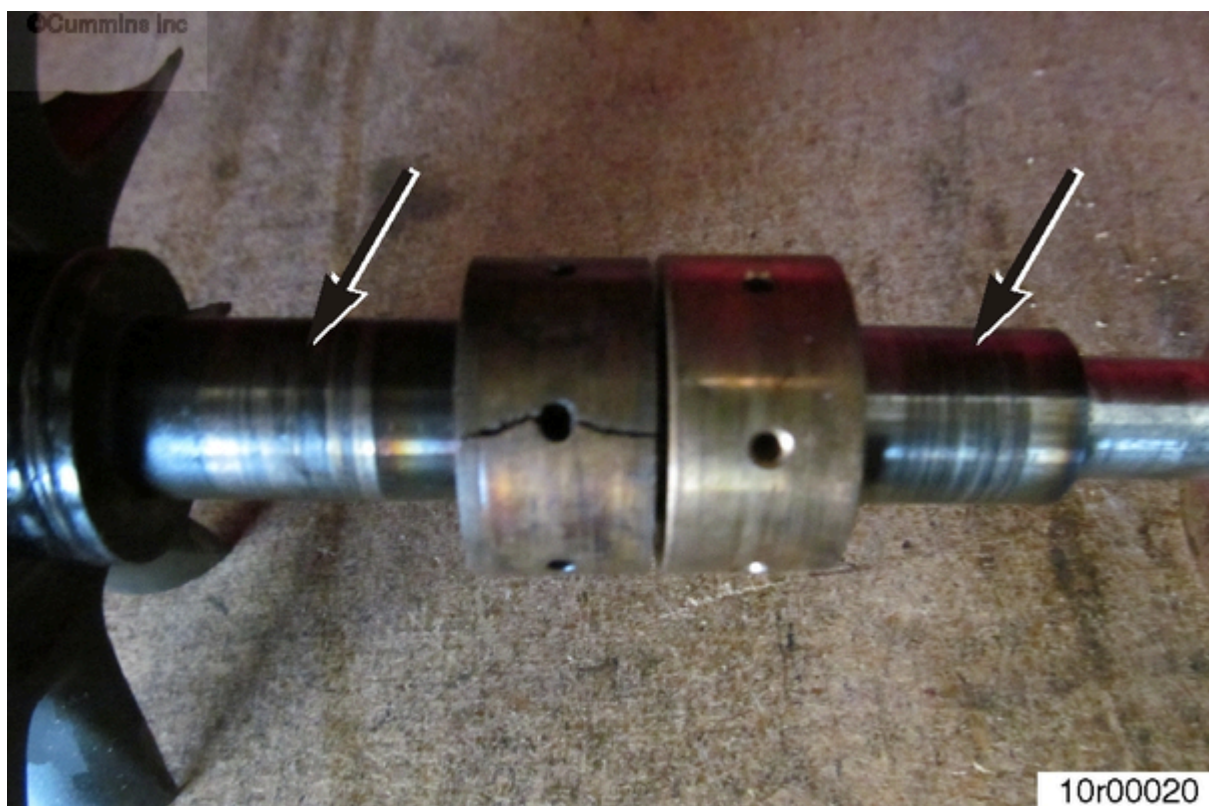


Figure 13: Damaged journal bearings.

Insufficient Lubrication:

Figure 12 and Figure 13 show damage from heat and material transfer due to high friction. Lack of lubrication creates high friction and heat leading to bearing wear.

Poor oil quality can also cause high friction and heat. The lubricating properties of oil can break down over time and cause insufficient lubrication. Contaminated oil from water/coolant can have the same effect.

Bearing failures can often be associated with the following characteristics:

- Axial and Radial Play are Out of Specification
- A Broken Shaft (When the wheel contacts its housing from bearing wear, the shaft can eventually fracture.)
- Noise (When the wheel begins to contact the housing from bearing wear, a high pitched noise can sometimes be heard.)
- Oil Leakage (When the axial and radial play are out of specification, the pressure seals can wear and cause oil leakage into the compressor and/or turbine housing.)

If a bearing failure is suspected, be sure to look into the cause of the bearing failure, as it could affect other engine components.

Oil Seal

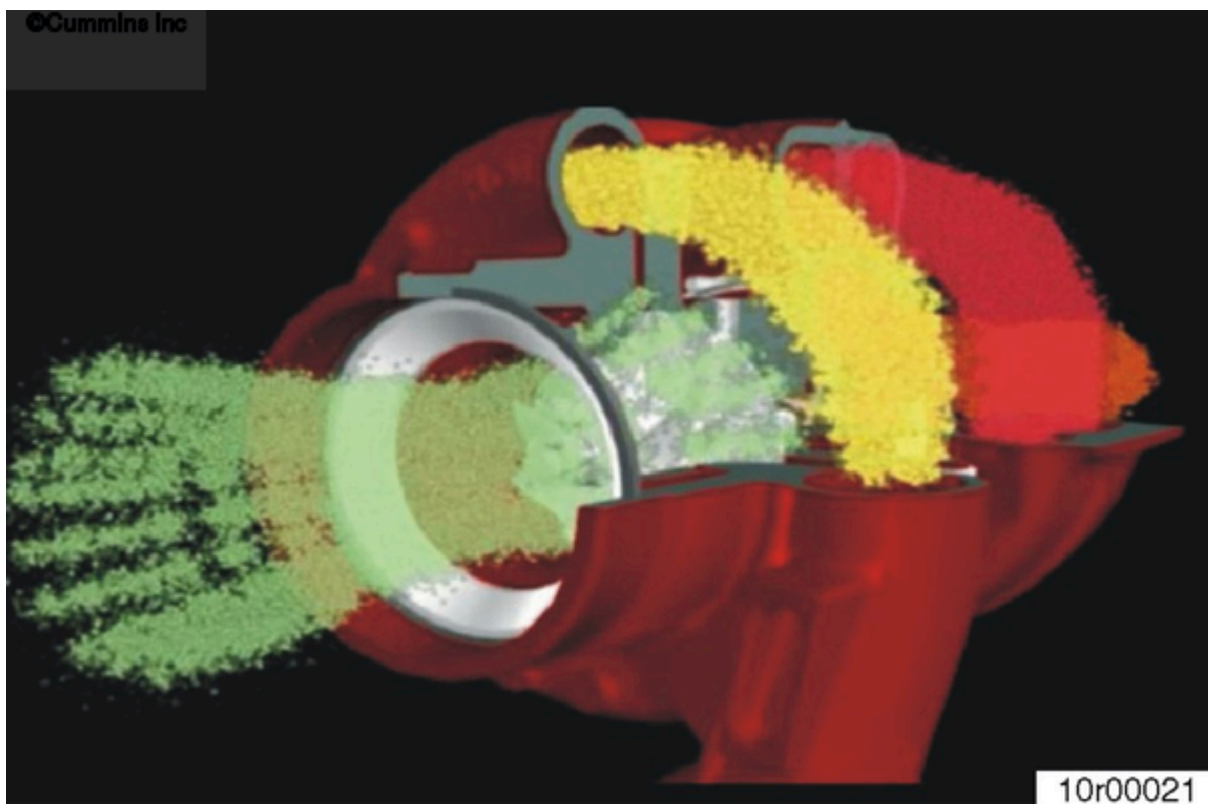


Figure 14: Pressure in the compressor and turbine housings.

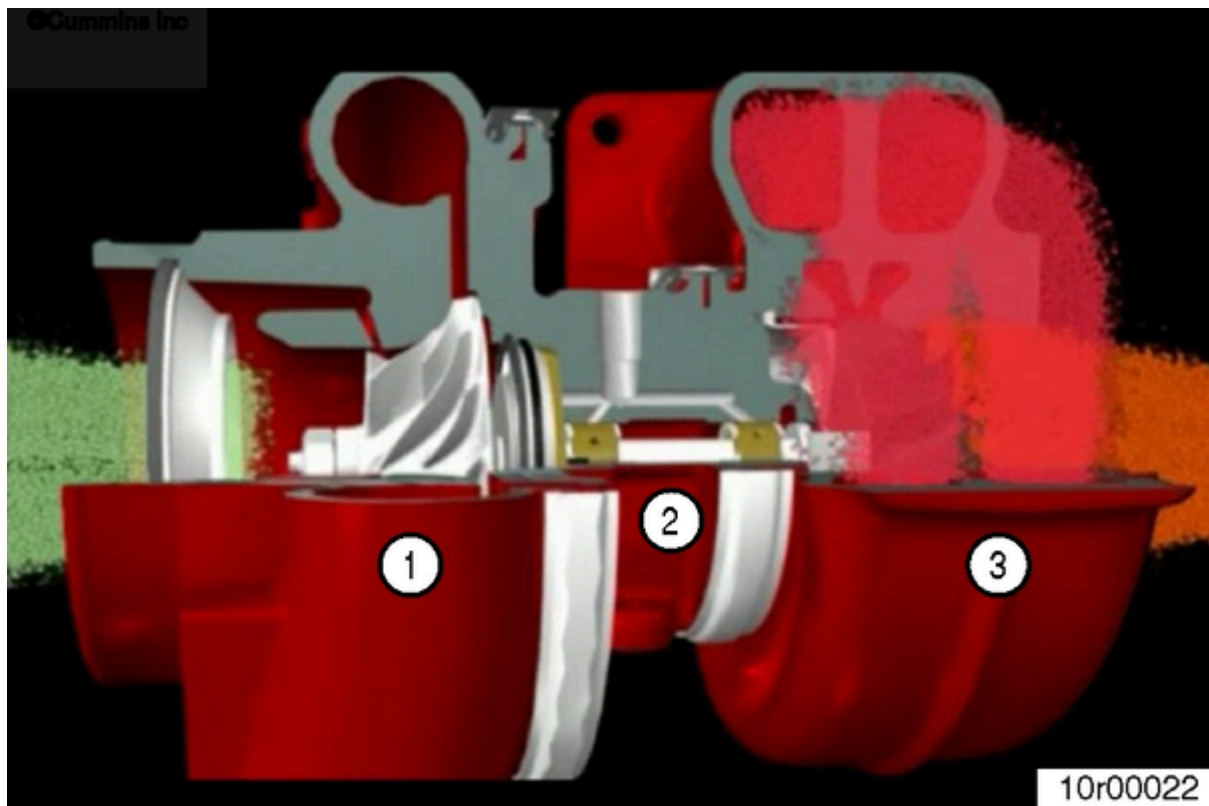


Figure 15: Compressor housing (1), bearing housing (2), and turbine housing (3).

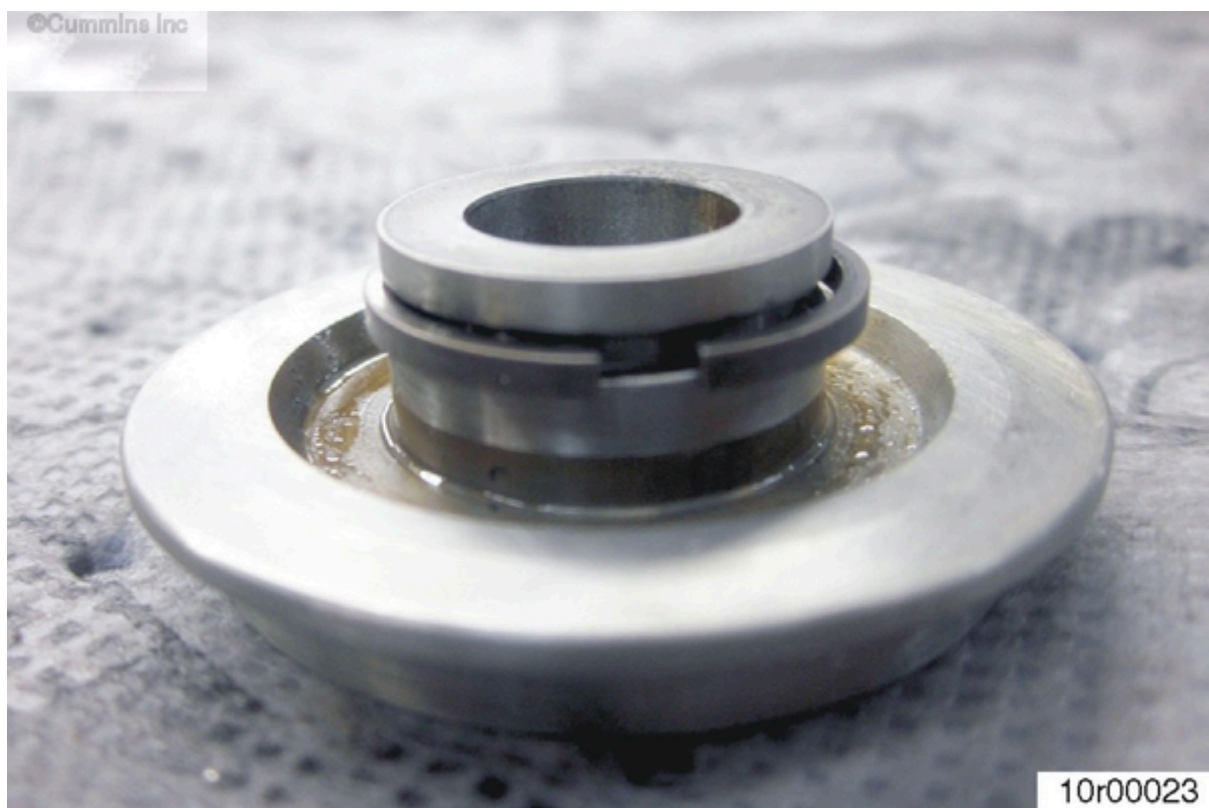
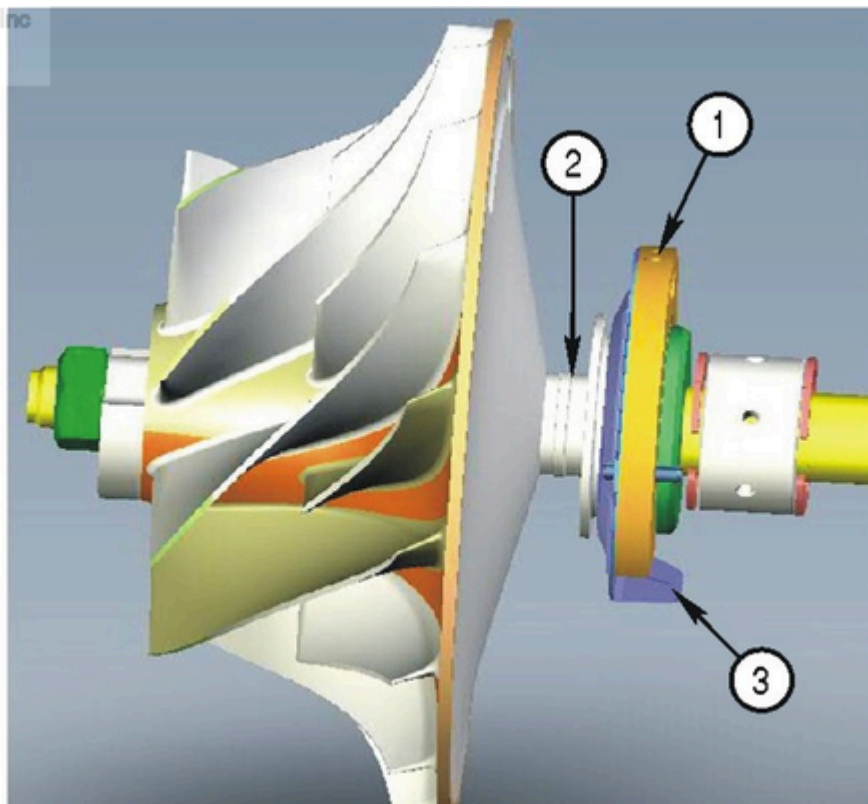


Figure 16: Oil Slinger with seal ring.



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Figure 17: Compressor impeller.

Figure 14 illustrates the pressure in the compressor and turbine housings. See Figure 15; The pressure in the compressor housing (1) and turbine housing (3) needs to be higher than the bearing housing (2) pressure to contain the lubricating oil in the bearing housing. The bearing housing pressure is nearly equal to the engine crankcase pressure.

Figure 16 shows an oil slinger with the compressor seal ring installed. The seal ring is a “piston style” seal that uses a positive pressure difference to seal oil. This is **not** a true oil seal.

The seal ring seals oil by having a higher pressure on the compressor side of the turbocharger than the bearing housing (center section).

Oil seeks low pressure and will stay in the bearing housing as long as the pressure is lower than the compressor housing pressure.

There are a number of outside factors that can allow compressor side pressure to drop or bearing housing pressure to rise. These factors are found to be external to the turbocharger.

See Figure 17; Oil is fed to the thrust bearing (1) to lubricate the oil slinger and thrust collar that mate against it. The seal ring (2) sits in the oil slinger. The oil slinger spins and directs oil into the oil baffle (3). Oil should **not** have contact with the seal ring. The oil slinger relies on turbocharger shaft rotation to keep oil away from the seals. The oil baffle is intended to direct oil to the turbocharger oil drain.

The seal ring is **not** a true oil seal. It is used, instead, as a pressure seal.

Common Factors for Turbocharger Oil Leakage

- Restricted Air Filters
 - A dirty air filter will lower pressure in the compressor housing, resulting in idle conditions allowing the bearing housing pressure to be higher than the compressor

housing.

- A non-approved air filter could be too restrictive and cause the same effect.

- Boost Leaks

- A boost leak can be from a torn hose, cracked charge-air cooler or aftercooler, improperly torqued clamps, or other conditions that can cause air to leak out of the air intake system.
- Instances where a boost hose pops off will act as a large boost leak.
- These cases cause a drop in pressure in the compressor housing and can lead to turbocharger oil leakage.

- High Crankcase Pressure

- Bearing housing pressure will roughly equal crankcase pressure. This pressure can rise from a restricted or collapsed crankcase filter. Crankcase pressure can also rise from a power cylinder issue.
- High amounts of blowby gas going up the oil drain will cause oil coming down the drain pipe to foam, preventing oil drainage and causing the turbocharger bearing housing to fill with oil.



Figure 18: Turbine wheel split ring oil seal.

Turbine Side Oil Leakage:

The turbine side of the turbocharger uses a split ring seal that works just as the compressor side.

Typical causes of noticeable oil leakage past this seal are from high crankcase pressure or a restricted oil drain.

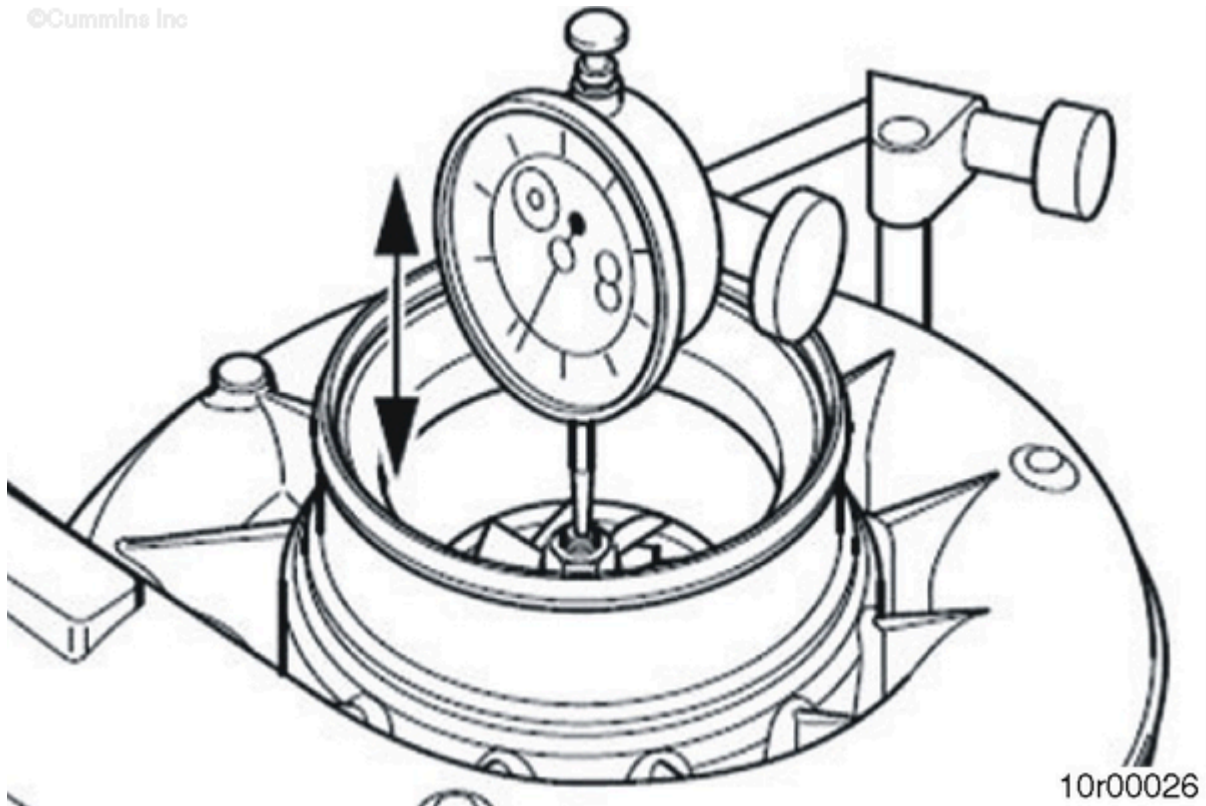


Figure 19: Measuring axial movement.

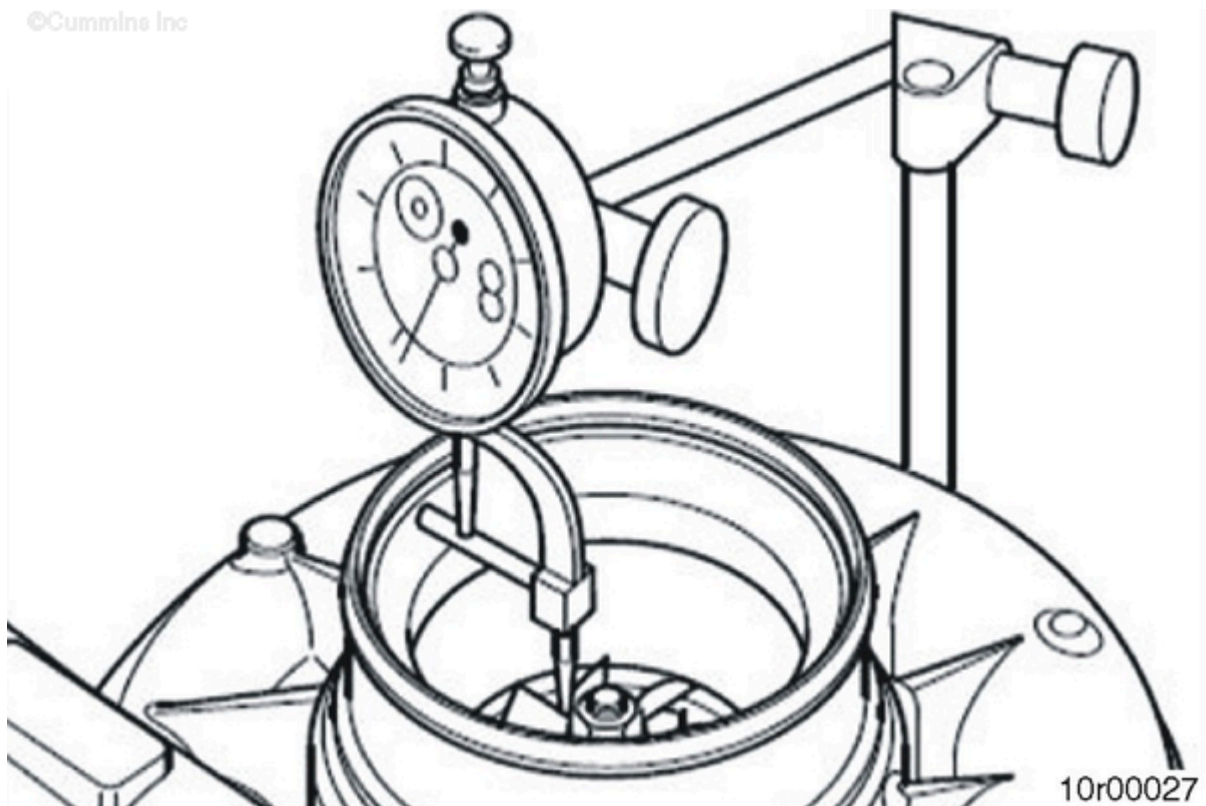


Figure 20: Measuring radial movement.

When the Turbocharger Should be Replaced:

- The compressor seal ring has **only** been found to malfunction if the rotor system has gone out of balance due to another turbocharger component failure.
- When diagnosing oil leakage into the air intake, if the axial and radial play of the turbocharger are measured to be within specification, the turbocharger should **not** be replaced.

Note : The cause of the oil leak **must** be found before allowing the engine back in service.

- The turbocharger axial and radial specifications can be found in Procedure 010-036 of the engine service manual.
- If the impeller wheel has **not** contacted the compressor housing, the axial and/or radial play are most likely to be within specification.

Turbocharger and Engine System Interaction

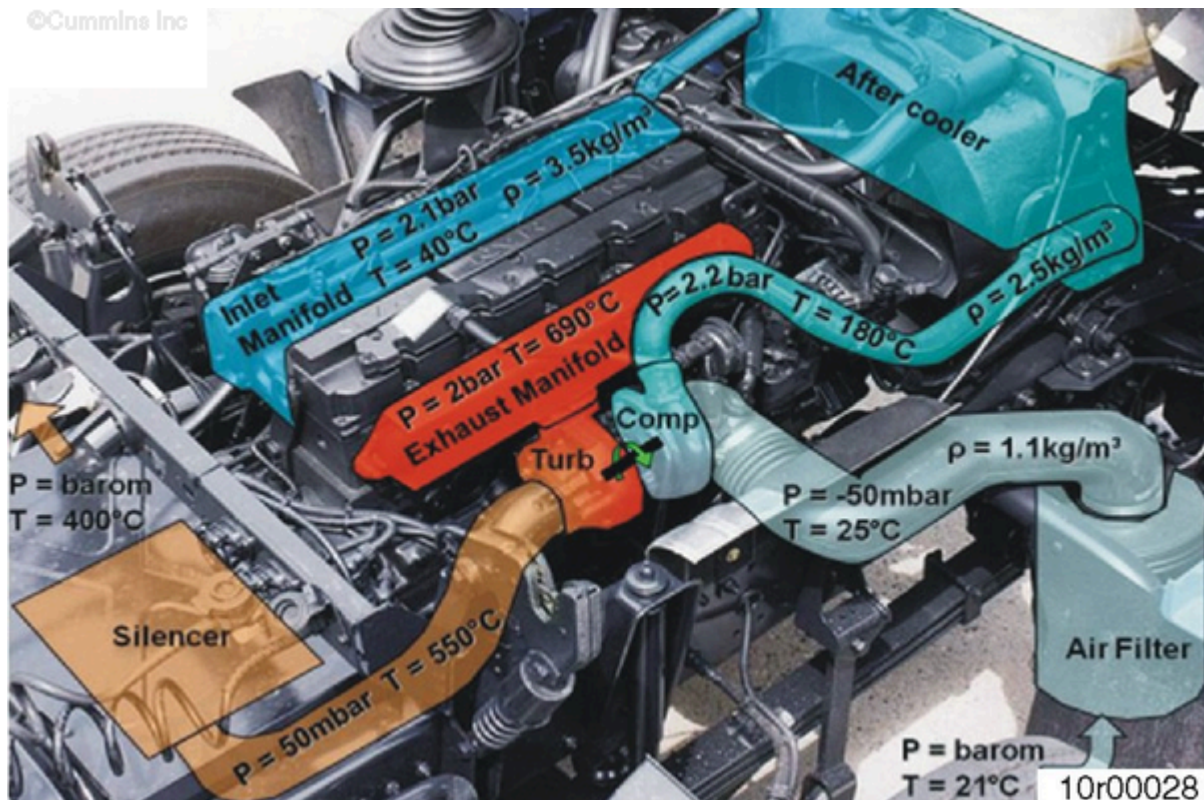


Figure 21



Figure 22: Damage to the turbine housing.



Figure 23: Damage to a heat shield.

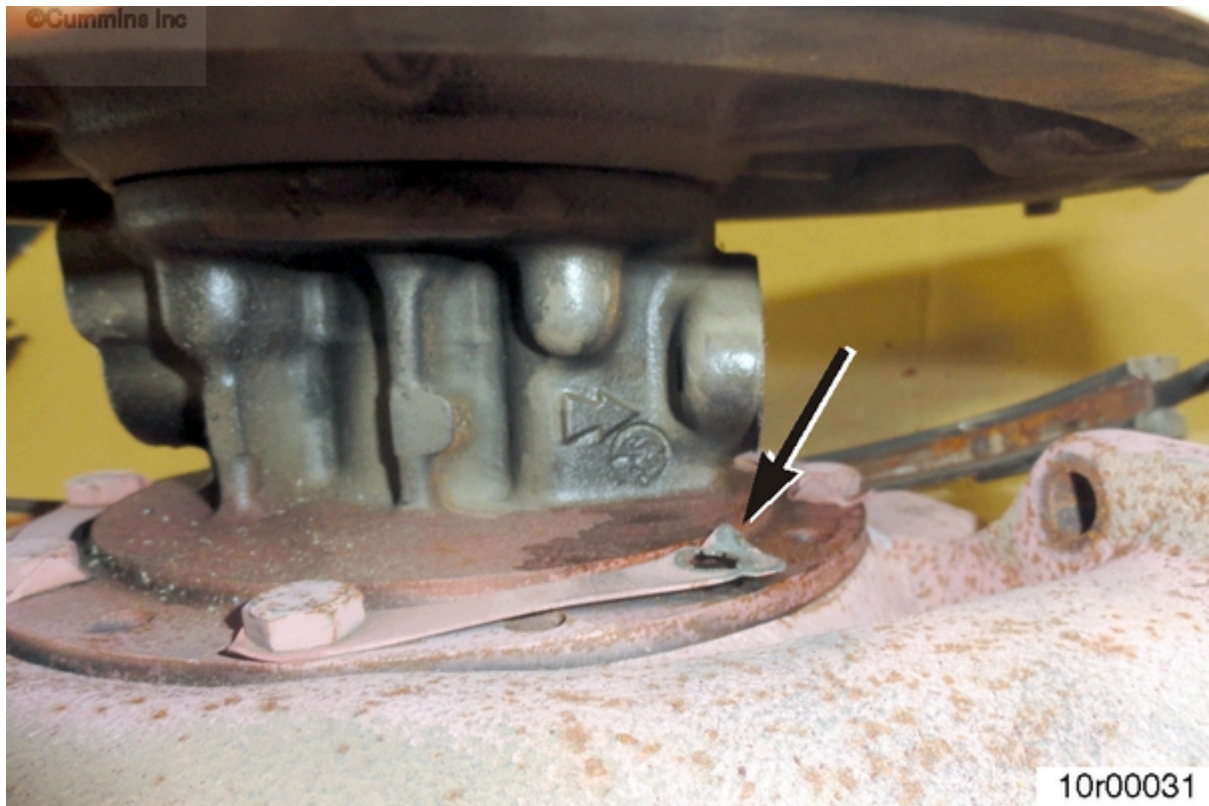


Figure 24: Capscrew missing from bearing housing-to-turbine housing connection.

High Exhaust Temperatures:

High exhaust temperatures can affect the turbocharger in a number of ways.

- The turbine journal bearing can fail due to overheated oil.
- The stamped steel heat shields can crack.
- The turbine to bearing housing capscrews can loosen/break due to thread relaxation.
- The turbine housing can crack externally.

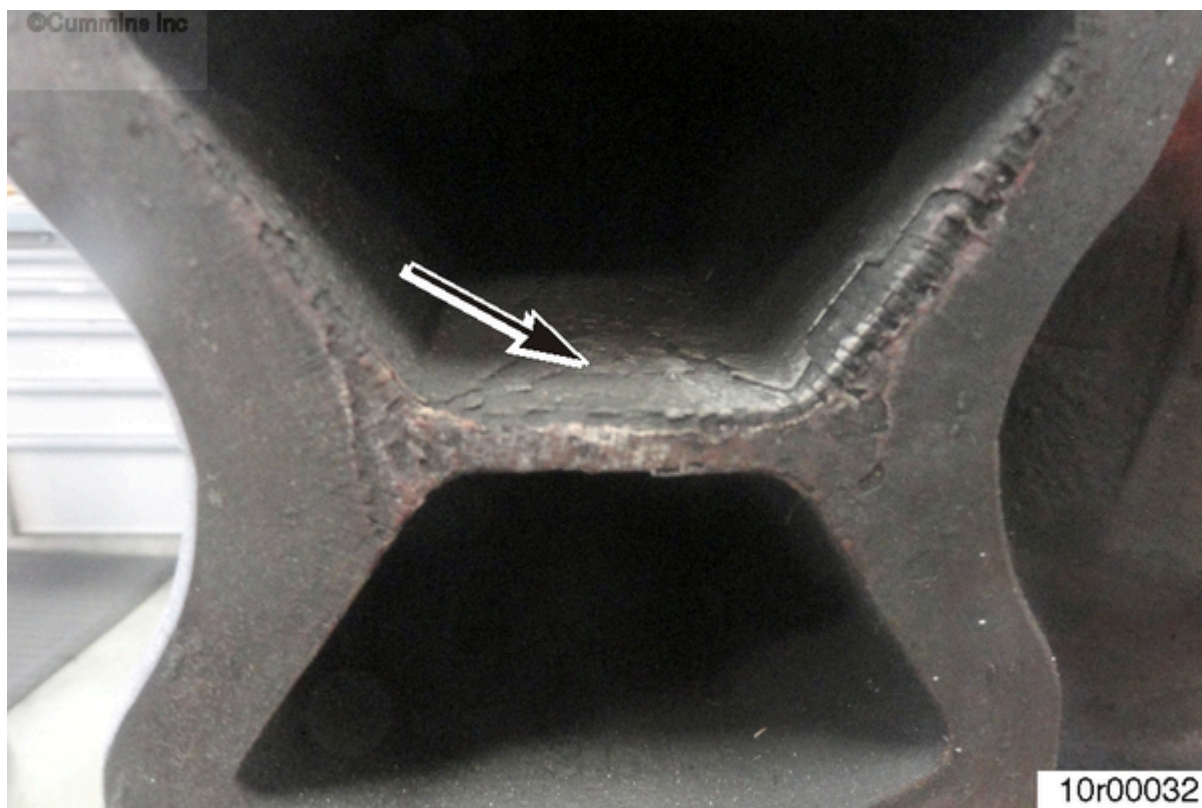


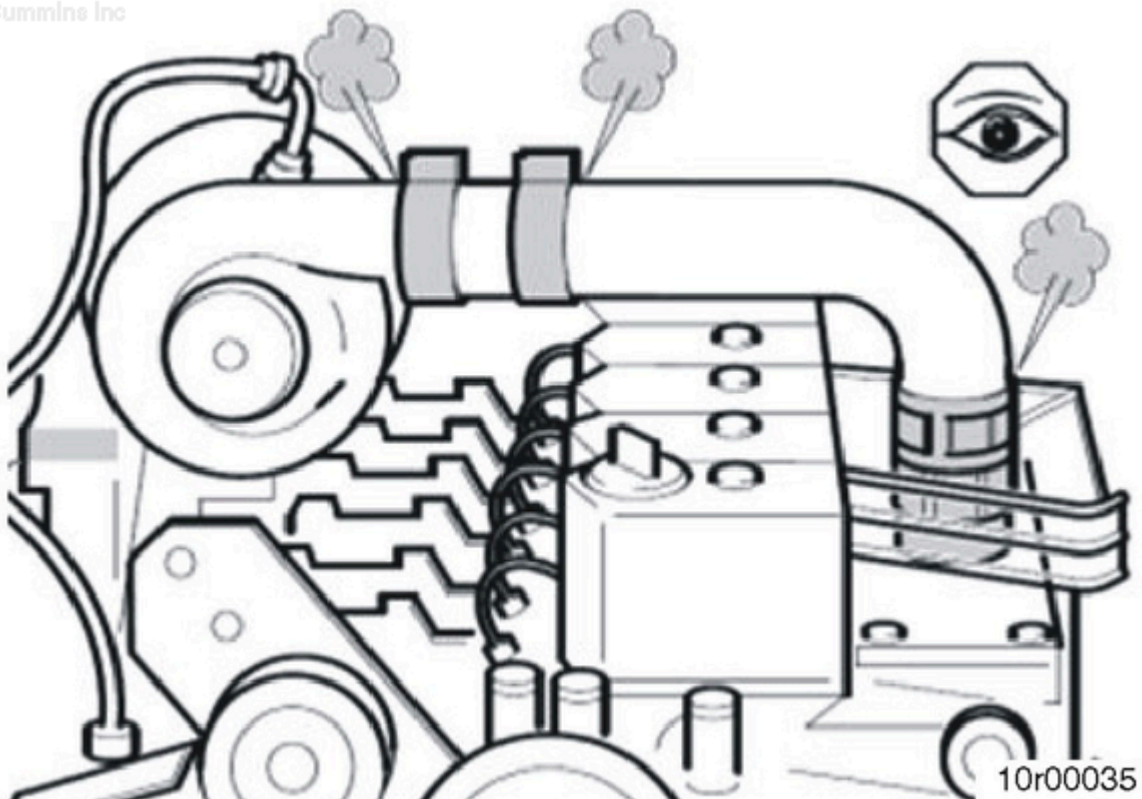
Figure 25: Scaling on turbine housing.



Figure 26: Scaling on exhaust manifold.

Evidence of high exhaust temperatures can be seen from scaling at the turbine housing and exhaust manifold joint (Figure 25). It can also sometimes be seen externally on the turbine housing and exhaust manifold (Figure 26). Material is seen removed from oxidation.

Exhaust temperatures on a diesel engine can rise from low intake manifold pressure, high intake manifold temperatures, overfueling, or high exhaust restriction.



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Figure 27: Boost leaks.



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Figure 28: Cooler plugged with debris.

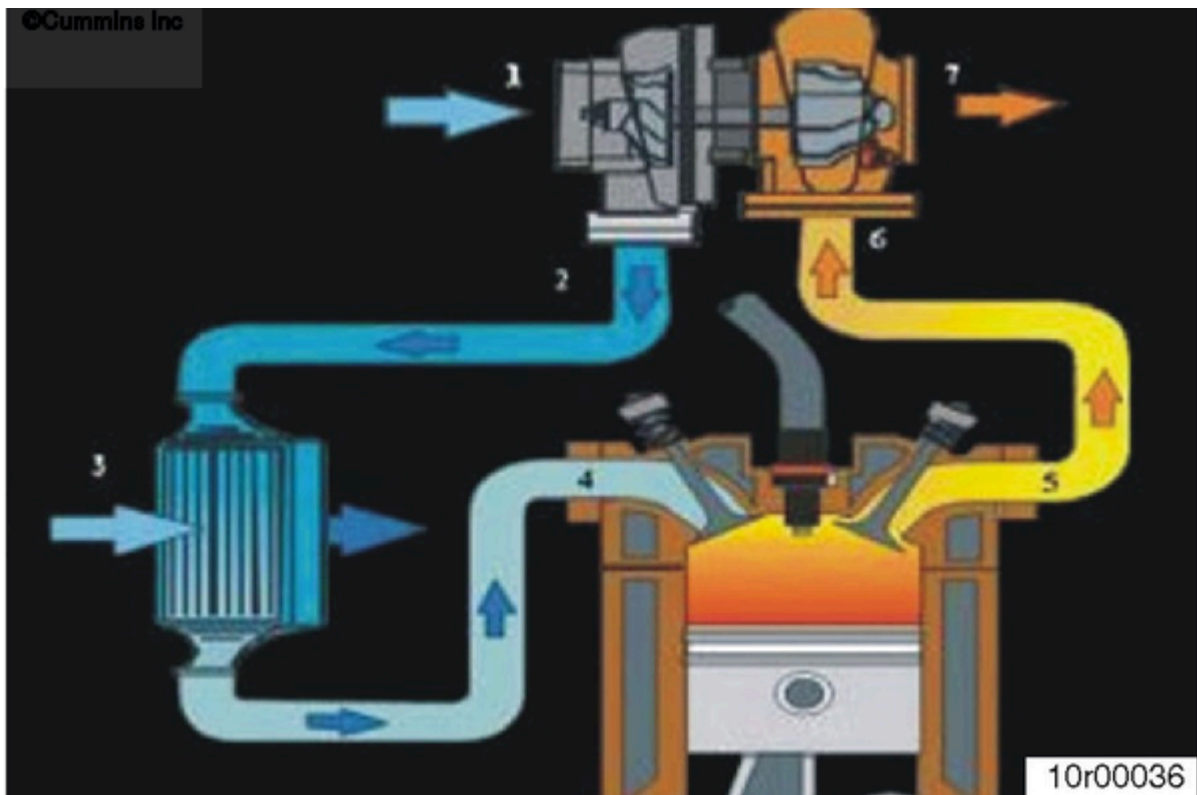


Figure 29

Low intake manifold pressure can cause a low air-to-fuel ratio. Overfueling conditions raise exhaust temperatures.

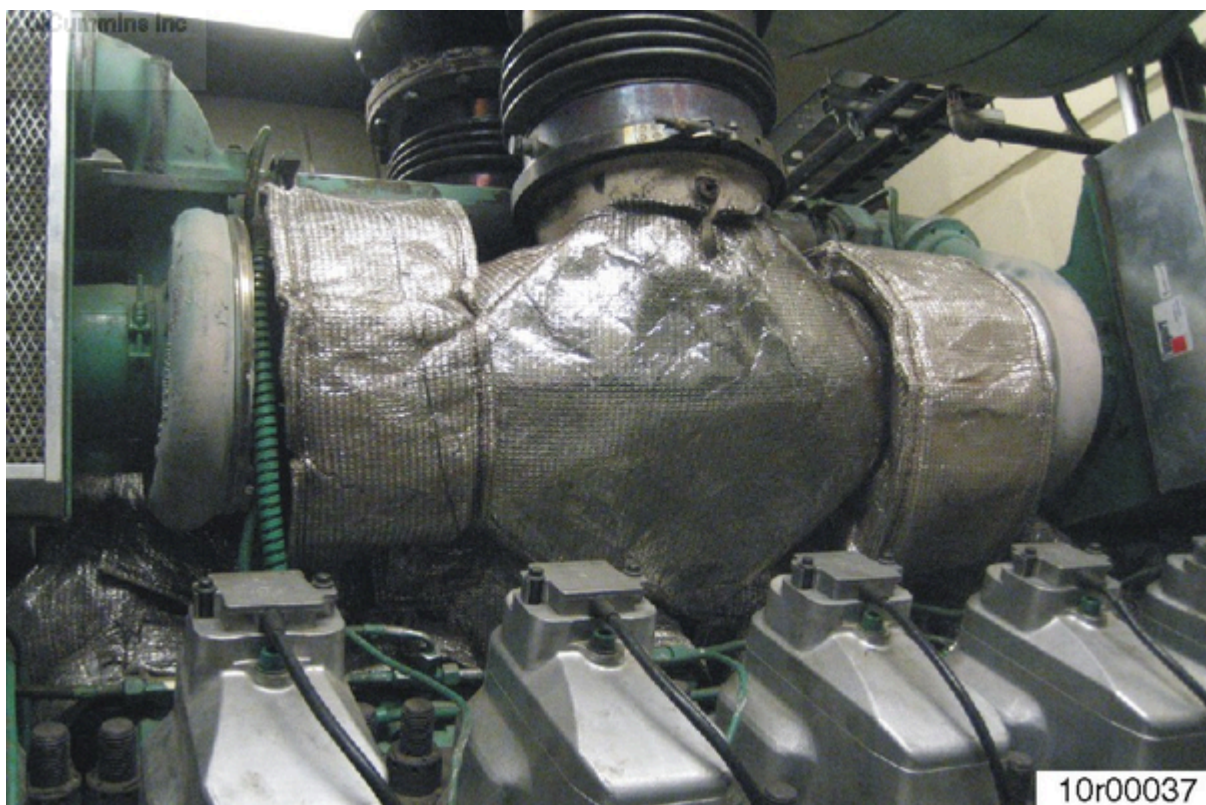


Figure 30: Turbocharger heat blanket.



Figure 31: Exhaust manifold heat blanket.

Heat blankets over the turbine housing and/or exhaust manifold can cause heat soak and increase exhaust temperatures. Heat blankets require approval for use in order to keep the exhaust temperatures in a safe range.



Figure 32: Fractured impeller wheel.



Figure 33: Collapsed intake air hose.

Turbocharger Overspeed

The turbocharger rotor system can reach speeds that exceed the capability of the components. The journal bearings can wear and the impeller can eventually fracture, as seen in Figure 32.

Figure 33 shows an air filtration setup that was too small for the engine and turbocharger. Part of the intake hose collapsed when the turbocharger was drawing more air than the filter would allow.

Common causes:

- High intake restriction or poor air filtration setup. The restriction of air to the turbocharger and engine causes the turbocharger speeds to increase as the engine requires more air.
- A boost leak can also increase rotor speeds as the engine requires more air.



Figure 34: Turbine wheel damage.

Foreign Object Damage (Exhaust Side)

Objects that enter the exhaust, such as a valve and piston material, or an injector tip, generally move at a very fast rate into the turbocharger. This can damage the blades of the turbine wheel.

Blade damage can be detected by looking into the turbine outlet with a flashlight.



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Figure 35: Impeller damage from small debris.



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Figure 36: Impeller damage from large debris.

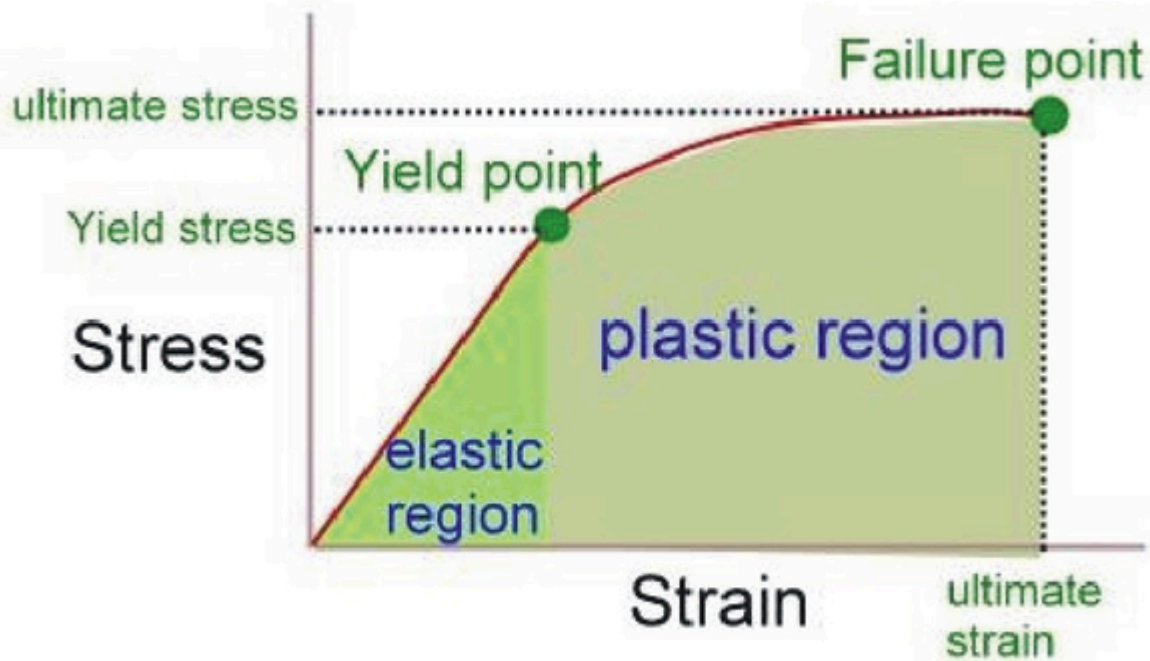
Foreign Object Damage (Compressor Side)

Objects such as bolts, nuts, sockets, hoses, or even shop towels have been known to get into the intake system and damage the turbocharger impeller.

Objects as small as weld splatter can cause some damage to the impeller blades (Figure 35), which could affect performance.

Larger objects entering the intake can cause a turbocharger failure, as seen in Figure 36.

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Figure 37: Wheel fatigue, stress vs. strain.



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Figure 38: Initiation point in failed impeller.



Figure 39: Initiation point in failed impeller.

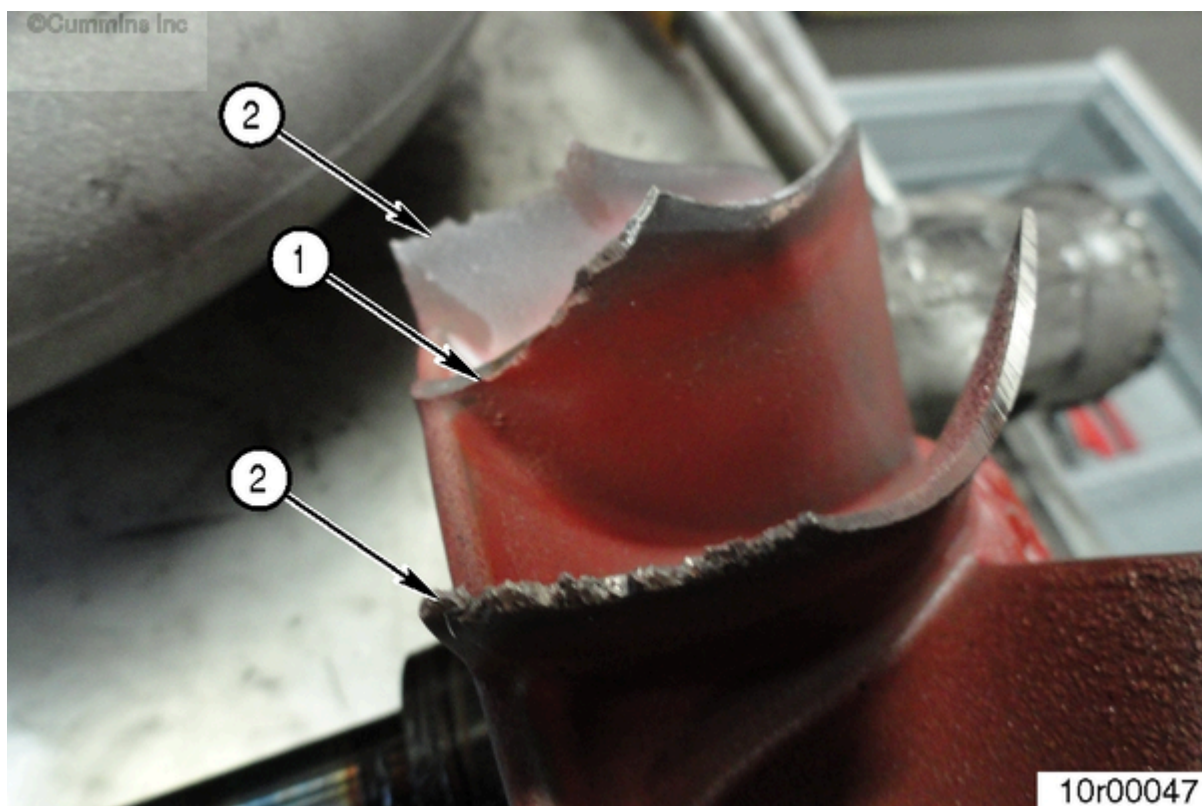


Figure 40: Damage from high cycle fatigue.

Wheel Fatigue

Low Cycle Fatigue

Figure 38 and Figure 39 show fatigue initiation points in impeller castings.

The stress and strain put on the impeller wheel will cause the wheel to fatigue over time. The fatigue life varies by wheel material, method of manufacturing, application, and other factors. Low cycle fatigue is most commonly found to occur on the impeller.

High Cycle Fatigue

Figure 40 shows a blade fractured from fatigue (1) is found to have a clean break in comparison to blades (2) damaged from impact. The blades labeled (2) were damaged from impact with pieces of the fatigued blade(s).

High cycle fatigue is most commonly found to occur on the turbine wheel. The fatigue occurs due to frequencies that can cause the blade to get excited and eventually fracture.

Summary

- A better understanding of what engine/system issues can have an effect on the turbocharger should help the understanding of the importance of certain maintenance practices.
- Turbochargers can fail in a number of ways, but the cause of the failure can often be related to an engine/system issue.
- Better understanding the cause of the turbocharger failure could help prevent further damage to the engine.

Additional Resources

Turbocharger Failure Diagnosis										
Possible Cause - Correction	Symptom									
	Engine Running Hot	Poor Transient Response	Smoke	Engine Lacks Power	Black Exhaust Smoke	Blue Exhaust Smoke	High Oil Consumption	Turbocharger Noisy	Cyclic Sound from Turbocharger	Oil Leak from Compressor Seal
Dirty air cleaner - Clean or replace element according to manufacturer's recommendations.	•	•	•	•	•	•	•			•
Restricted compressor intake duct - Remove restriction or replace damaged parts as required.	•	•	•		•	•	•	•	•	•
Restricted air duct from compressor to intake manifold - Replace seals, gaskets or tighten fasteners as required	•	•		•	•			•		
Restricted intake manifold - Refer to engine manufacturer's manual and remove restriction	•	•		•	•			•		
Air leak in feed from air cleaner to compressor - Replace seals, gaskets or tighten fasteners as required								•		
Air leak in feed from compressor to intake manifold - Replace seals, gaskets or tighten fasteners as required	•	•	•	•	•	•	•	•		
Air leak between intake manifold and engine - Refer to engine manufacturer's manual and replace gaskets or tighten fasteners as required	•		•	•	•	•	•	•		
Foreign object in exhaust manifold (from engine) - Refer to engine manufacturer's manual and remove obstruction				•	•	•	•	•		•
Restricted exhaust system - Remove restriction or replace damaged parts as required	•			•	•					•
Exhaust manifold cracked, gaskets blown or missing - Refer to engine manufacturer's manual and replace gaskets or damaged parts as required		•	•	•	•			•		
Gas leak at turbine inlet/exhaust manifold joint - Replace gasket or tighten fasteners as required		•	•	•	•			•		
Gas leak in ducting after turbine outlet - Refer to engine manufacturer's manual and repair leak		•						•		

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Figure 41: Turbocharger Troubleshooting Chart, Part 1.

Use the charts in Figure 41 and Figure 42 to verify the problem before replacing the turbocharger.

Turbocharger Failure Diagnosis										
Possible Cause - Correction	Symptom									
	Engine Running Hot	Poor Transient Response	Smoke	Engine Lacks Power	Black Exhaust Smoke	Blue Exhaust Smoke	High Oil Consumption	Turbocharger Noisy	Cyclic Sound from Turbocharger	Oil Leak from Compressor Seal
Restricted turbocharger oil drain line - Remove restriction or replace damaged parts as required						•	•			•
Restricted engine crankcase breather - Refer to engine manufacturer's manual, clear restriction						•	•			•
Turbocharger bearing housing sludged or coked - Change engine oil and oil filter, overhaul or replace turbocharger as required						•	•			•
Fuel injection pump or fuel injectors incorrectly set - Refer to engine manufacturer's manual and replace or adjust faulty components as required		•	•	•	•					
Engine valve timing incorrect - Refer to engine manufacturer's manual for correct settings and adjust as required				•	•					
Worn engine piston rings or liners - Refer to engine manufacturer's manual and repair as required				•	•	•	•			•
Burnt valves and/or pistons - Refer to engine manufacturer's manual and repair as required				•	•	•	•			•
Excessive dirt build-up on impeller and/or diffuser vanes - Clean in accordance with the appropriate Holset repair manual				•	•	•	•	•	•	•
Turbocharger damaged - Find and correct cause of failure, repair or replace turbocharger as necessary				•	•	•	•			•
Failed diaphragm - Replace using correct Actuator Service Kit	•							•		
Siezed valve - Free valve/replace complete turbine housing sub-assembly	•	•								
Leaking hose - Replace hose and clips	•							•		
Wastegate mechanism set incorrectly - Contact nearest authorised Holset distributor for correct setting details	•	•	•	•				•		

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Figure 42: Turbocharger Troubleshooting Chart, Part 2.

Use the charts in Figure 41 and Figure 42 to verify the problem before replacing the turbocharger.

Document History

Date	Details
2013-3-11	Module Created

Last Modified: 04-Jun-2013